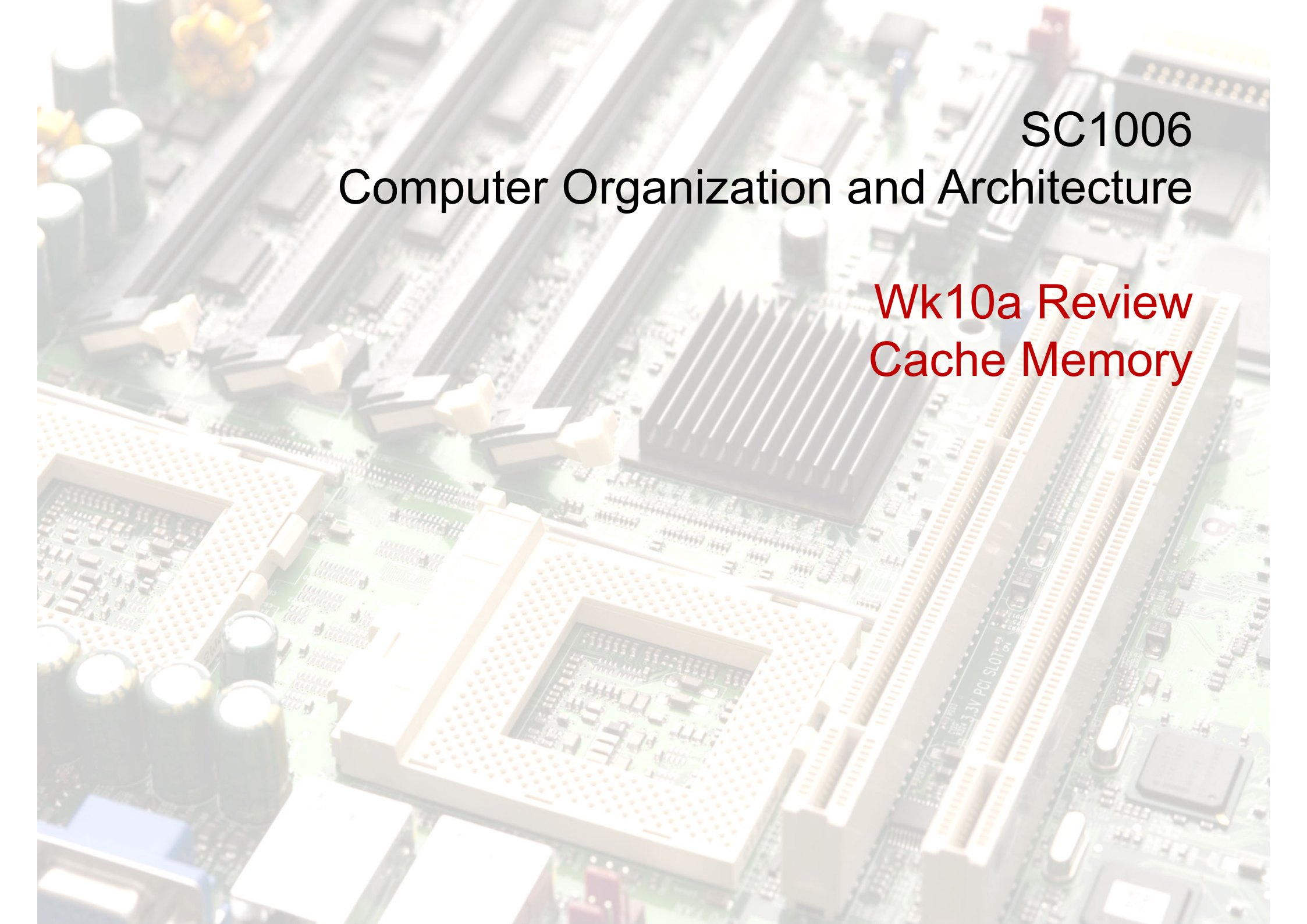




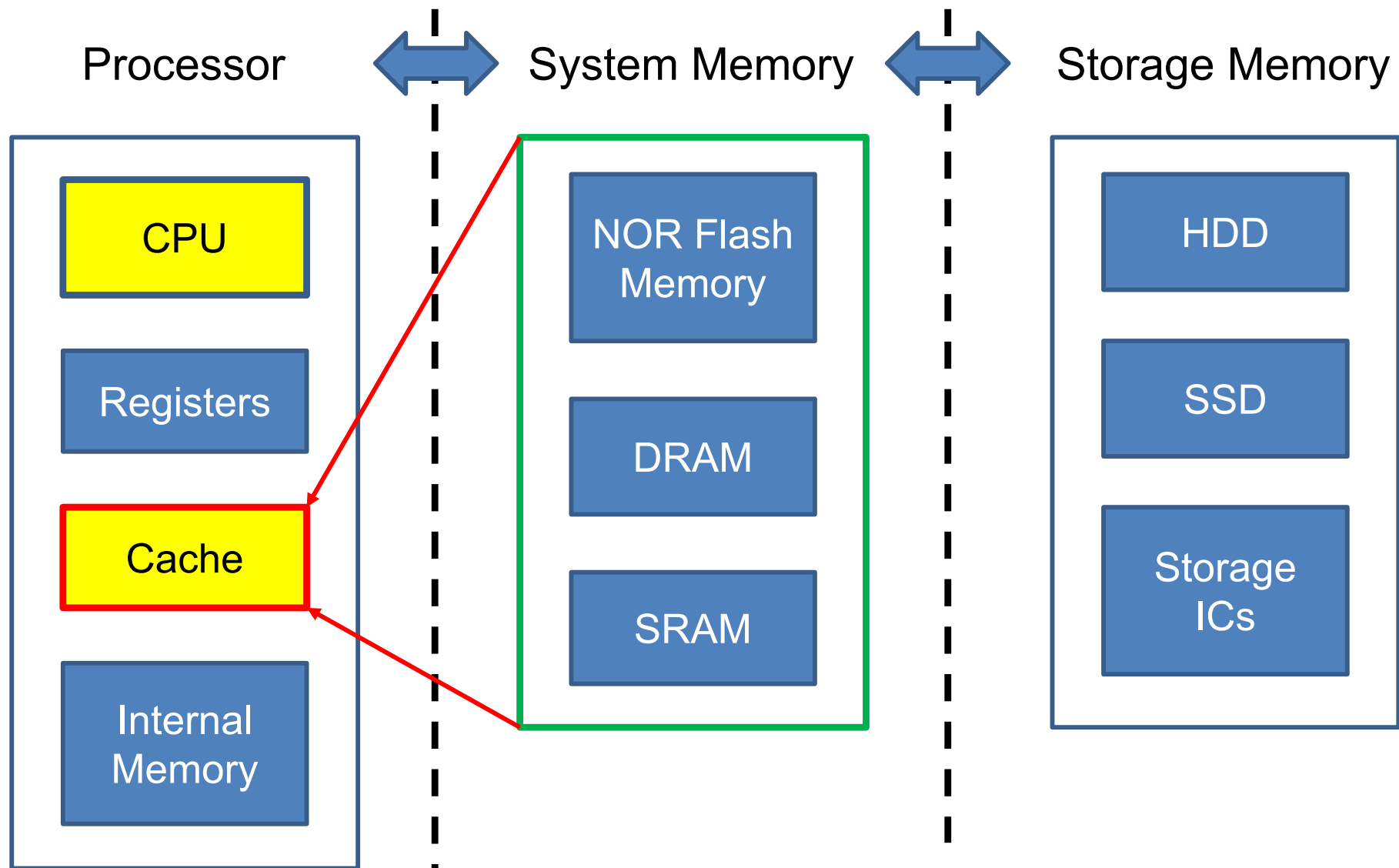
SC1006

Computer Organization and Architecture

Wk10a Review
Cache Memory

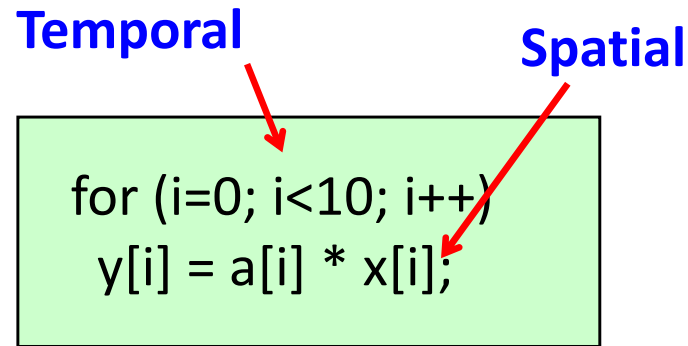
CACHE

Computer Memory (Programmer's View)



Increasing Memory Speed and Memory Cost

Principle of Locality



- The **principle of locality** tells us that once a byte in a program is accessed, it is likely a **nearby data element** will be needed soon.
- There are two principle of locality governing this behaviour
- Locality of **Space** (or Spatial Locality)
 - **Code/Data that is nearby each other** is likely to be accessed together
 - Transfer of data between System memory and cache is done in blocks to leverage on this behaviour
- Locality of **Time** (or Temporal Locality)
 - **Recently accessed code/data** is more likely to be accessed again
 - Used to decide which item to replace in the Cache

thinking questions toward understanding...

Suppose that the cache management is with write back policy. What happens when a cache block is selected for replacement?



thinking questions toward understanding...

Does additional cache memory
always increase system performance?



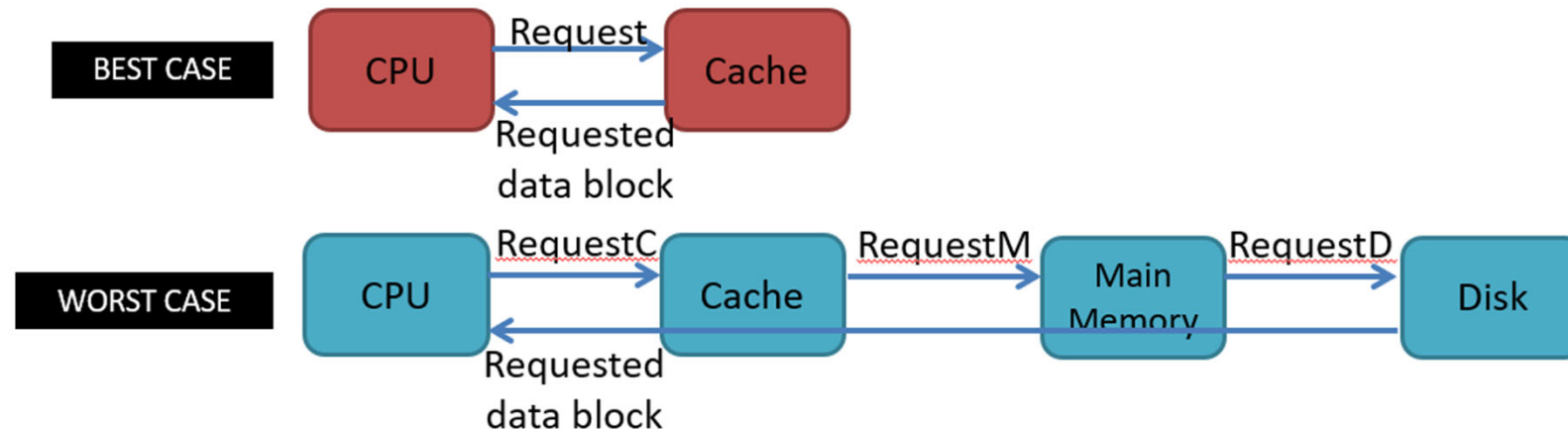
Online Quiz Review

****Which of the following is/are true for Cache Memory?**

- ✓ A. Build with Fast Memory
- ✓ B. Contain a subset of data/code in main and storage memory
- ✓ C. Contain frequently used data/code
- ✓ D. Transfers to cache are in blocks instead of single word.

When CPU needs some data/code, which module does it check first for the information required?

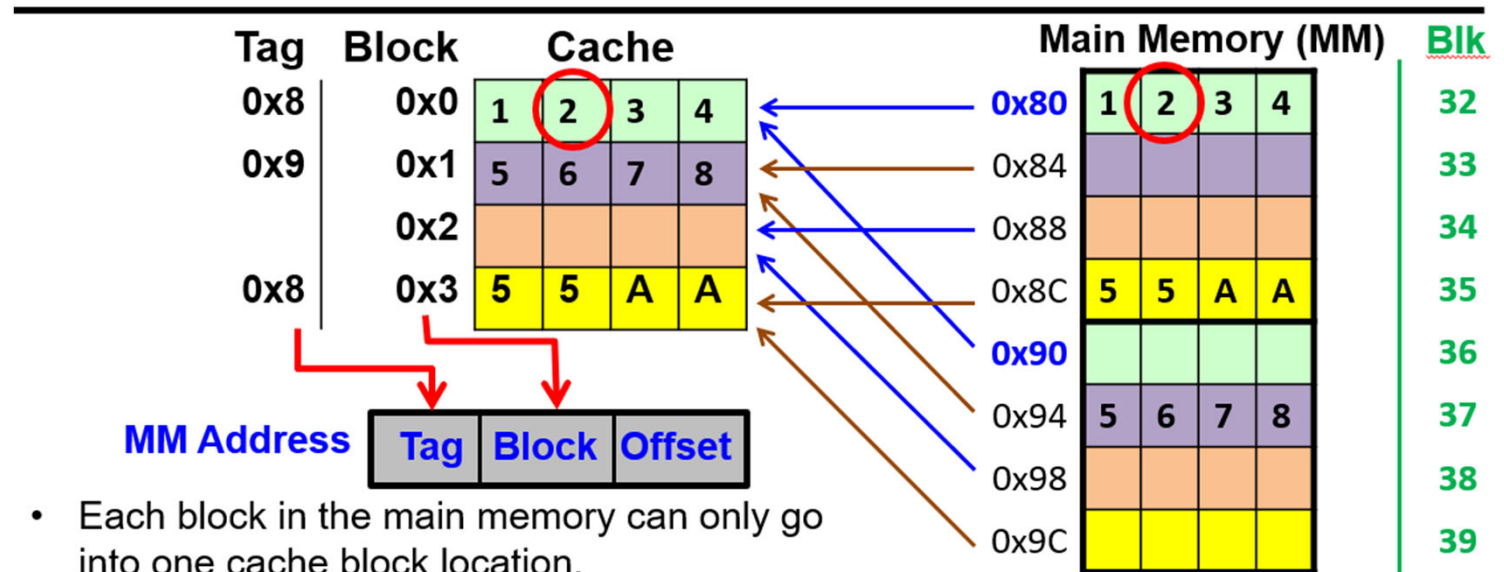
- A. Main Memory
- B. Storage Memory
- ✓ C. Cache
- D. Virtual Memory



How many location(s) within the cache can a main memory block be mapped to in a direct mapped cache?

- ✓ A. 1
- B. 2
- C. 4
- D. Any block within the cache

Direct Mapped Cache



Does Direct Mapped Cache require a cache replacement policy?

- ✓ A. No
- B. Yes

Cache replacement policy is employed when the cache needs to decide which particular cache block among all cache blocks to evict in order to free up space for new data from main memory to be transferred into the cache. That implicitly means that the main memory block could potentially be transferred to one or more cache block location, in general.

However, in direct mapped cache, each main memory can only map to one and only one cache block location. Meaning there is no need to apply any cache replacement policy algorithm to decide where a particular main memory block should be transferred to.

Given a MM size of 64KByte, Cache size of 256Bytes and block size of 16 bytes. Direct Mapped Cache. What is the format for Tag:Block:Offset of the cache?

- A. 4:4:8
- ✓ B. 8:4:4
- C. 4:8:4
- D. None of the above

64K => 16bits address

Cache Block size 16bytes => 4bits offset

Cache Size 256bytes => 16 blocks => 4bit block index

=> 8 bit Tag

What is the Tag value of MM block with starting address 0x20. Assume a Direct-Mapped Cache with 4 blocks, block size 4 bytes and 0xFF MM address range.

- ✓ A. 0x2
- B. 0x10
- C. 0x0
- D. 0x20

0xFF => 8 bits address
=> 4:2:2
0x20 = 0010 0000b
=> TAG = 0010b = 0x2

Which of the following statement(s) illustrate the principle of spatial locality?

- A. Code in a loop gets executed multiple times
- ✓ B. Code is often executed sequentially
- ✓ C. Data in an array are accessed one after another typically
- ✓ D. Variables used within a function are typically assigned near to each other by the compiler

[B] Code executed sequentially means that the code is fetched from sequential addresses i.e. location that are near to each other.

[C] Data in an array are allocated in sequential memories by the compiler.

[D] Variables within a function are typically accessed together or in close vicinity.

[A] Code executed multiple times is a form of **temporal locality**.

Which of the following statements illustrate the principle of temporal locality?

- ✓ A. Code in a loop gets executed multiple times
- B. Code is often executed sequentially
- C. Data in an array are accessed one after another typically
- ✓ D. Variables within a function are typically used multiple times in the function

[A], [D]. Code/Data being used multiple times repeatedly is a form of temporal locality.

Notes: [D] is different from [D] from the previous Q.

(a function can be with both spatial locality and temporal locality if the function is called repeatedly.)

[B], [C]. These are spatial locality, as explained in the previous question.

Which principle of locality is the LRU replacement policy based on?

- A. Locality of space
- ✓ B. Locality of time

LRU (**least recently used**) replacement policy replaces the cache block that hasn't been used for the longest period of time. This is based on the locality of time.

Since the particular cache block hasn't been used for a while, that means the chances of it being required again is not high compared to the rest of the blocks in the cache, hence it is chosen to be purged/evicted.

What is cache coherency issue?

- ✓ A. Content in the cache is different from the content of corresponding locations in the main memory
- B. Too many access to the cache
- C. Cache access not in sync with memory access
- D. Cache data corrupted

Cache coherency refers to the situation where the copy of data in the cache is different from its corresponding copy in the main memory.

How does cache coherency issue occur?

- A. Too many reads from Cache.
- ✓ B. CPU writes to data in the cache but the copy of data in the main memory is not updated accordingly
- C. Using Write-Through Policy
- D. Using LRU Cache replacement policy

A similar question as the previous one.

Cache coherency refers to the situation where the copy of data in the cache is different from its corresponding copy in the main memory.

This typically happens when CPU writes to the cache while the copy in the main memory hasn't been updated (yet).

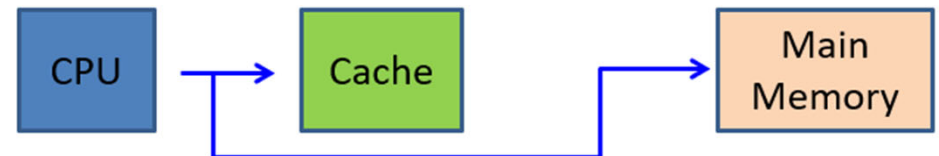
What is the advantage that Write-Through policy has over Write-Back one?

- A. Less traffic on system bus
- B. Faster write throughput
- ✓ C. No cache coherency issue
- D. No additional main memory write needed for write to cached data

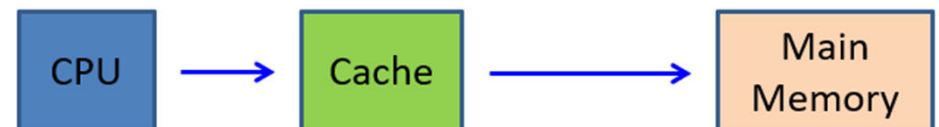
With Write-Through policy, all writes to the cache is accompanied by an update to the corresponding main memory location.

Hence, there will be no cache coherency issue.

- **Write through** updates cache and main memory **simultaneously** on every write.



- **Write back** (also called copyback) updates memory **only** when the block is selected for replacement.



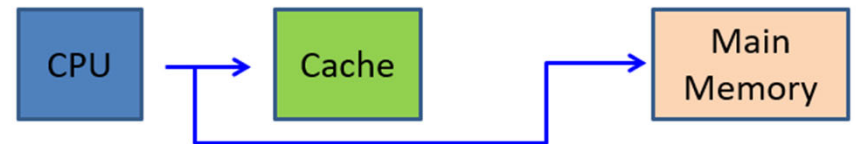
What is the disadvantage that Write-Through policy has over Write-Back one?

- ✓ A. Generate more traffic on system bus
- B. More complex cache data manipulation
- C. Worsen the cache coherency issue
- D. Delayed updating of cached data

With Write-Through policy, all writes to the cache is accompanied by an update to the corresponding main memory location.

This memory write operation requires use of the system bus so will inevitably generate more traffic on the system bus.

- **Write through** updates cache and main memory **simultaneously** on every write.



- **Write back** (also called copyback) updates memory **only** when the block is selected for replacement.

